AI-ENHANCED SMART GESTURE COMMUNICATION GLOVE FOR ASSISTIVE COMMUNICATION

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ABSTRACT:

Communication barriers can affect individuals with speech and hearing impairments, particularly when communicating with people who are unfamiliar with sign language. This paper proposes an AI-enhanced smart gesture communication glove intended to support accessible communication by translating predefined hand gestures into text and, potentially, speech. The proposed wearable system integrates five flex sensors to measure finger bending and an MPU6050 inertial measurement unit (IMU) to capture hand orientation and motion. An ESP32 microcontroller is intended to collect and process the sensor readings. These readings will be used to develop a gesture-recognition pipeline based on a Random Forest classification model, covering selected Indian Sign Language (ISL) signs as well as general hand gestures. The recognized gesture can be mapped to a corresponding text label and displayed on a 0.96-inch OLED screen, while voice output is considered as an additional accessibility feature. By combining finger-flexion measurements with motion and orientation data, the design aims to distinguish a broader range of gestures in a portable wearable form. The project is currently at the proposed design stage; dataset collection, model training, hardware integration, and systematic evaluation will be required to assess recognition performance and usability. The proposed system seeks to explore an affordable embedded-AI approach to assistive communication and provide a foundation for future expansion of the gesture vocabulary and real-world testing.

Keywords: Assistive Communication; Gesture Recognition; Indian Sign Language; Flex Sensors; ESP32; Random Forest; Wearable Computing.

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